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(An Autonomous Institute, with NAAC "A" Grade)

Affiliated to DBATU, RTMNU & MSBTE Mumbai

Department of Computer Science & Engineering

"A Place to Learn, A Chance to Grow"

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VISION

To be recognized for excellent engineering, developing global leaders both in educational and research in the domain of computer science and wireless engineering.

MISSION

1. To create self-learning environment by facilitating leadership qualities, team spirit and ethical responsibilities.
2. To improve department-industry collaboration, interaction with professional society through technical knowledge and internship program.
3. To promote research and development with current techniques through well qualified resources in the area of computer science and wireless engineering.

Practical No. 01

Aim: To study of Python Basics Libraries such as Statistics, Math, Numpy, Scipy and Matplotlib.

Software Required: Python IDE, Anaconda, VS Code, Jupyter Nootbook, Google Colab.

Theory:

Python standard library

The Python Standard Library contains the exact syntax, semantics, and tokens of Python. It contains built-in modules that provide access to basic system functionality like I/O and some other core modules. Most of the Python Libraries are written in the C programming language. The Python standard library consists of more than 200 core modules. All these work together to make Python a high-level programming language. Python Standard Library plays a very important role. Without it, the programmers can't have access to the functionalities of Python. But other than this, several other libraries in Python that make a programmer's life easier. Let's have a look at some of the commonly used libraries:

1) Statistics:

- The statistics library provides functions for calculating mathematical statistics of numeric data.
- Statistics is the branch of mathematics concerned with the collection, analysis, interpretation, and presentation of data.
- In Python, statistical analysis can be performed using various libraries that make it easier to work with both small and large datasets.
- These libraries provide functions for descriptive statistics, inferential statistics, and data modeling.
- This module provides basic statistical functions for calculating measures of central tendency (mean, median, mode) and measures of dispersion (variance, standard deviation). It is suitable for smaller datasets and quick calculations without external dependencies.

a. Mean

It is the sum of observations divided by the total number of observations. It is also defined as average which is the sum divided by count.

$$Mean(\bar{x}) = \frac{\sum x}{n}$$

b. Median

It is the middle value of the data set. It splits the data into two halves. If the number of elements in the data set is odd then the center element is the median and if it is even then the median would be the average of two central elements. it first sorts the data and then performs the median operation

For Odd Numbers:

$$\frac{n+1}{2}$$

For Even Numbers:

$$\frac{\frac{n}{2} + (\frac{n}{2} + 1)}{2}$$

c. Mode

It is the value that has the highest frequency in the given data set. The data set may have no mode if the frequency of all data points is the same. Also, we can have more than one mode if we encounter two or more data points having the same frequency.

2) Math:

- Math Module consists of mathematical functions and constants. It is a built-in module made for mathematical tasks.
- The math module provides the math functions to deal with basic operations such as addition (+), subtraction (-), multiplication (*), division (/), and advanced operations like trigonometric, logarithmic, and exponential functions.
- It is suitable for general mathematical calculations and is always available without needing to be installed separately.
- **Example:** Importing Math Module

```
import math
```

3) Numpy:

- NumpyThe numpy library is used for working with arrays and provides a large set of mathematical functions to operate on these arrays.
- NumPy is Python's fundamental package for scientific computing. It is a library that provides a multidimensional array object, various derived objects (such as masked arrays and matrices), and an assortment of routines for fast operations on arrays, including mathematical, logical, shape manipulation, sorting, selecting, I/O, discrete Fourier transforms, basic linear algebra, basic statistical operations, random simulation and much more.
- NumPy is used at the core of many popular packages in the world of Data Science and machine learning.

Installing NumPy in Python

To begin using NumPy, you need to install it first. This can be done through pip command:

pip install numpy

Once installed, import the library with the alias `np`

```
import numpy as np
```

4) scipy:

- The scipy library builds on numpy and provides a large number of functions that operate on numpy arrays and are useful for scientific and technical computing.
- SciPy (pronounced "Sigh Pie") is an open-source Python library designed for scientific and technical computing. It is built upon the NumPy library, which provides the fundamental array object and basic numerical operations. SciPy extends NumPy's capabilities by offering a wide range of high-level functions for various scientific and engineering tasks.
- SciPy is an interactive Python session used as a data-processing library that is made to compete with its rivalries such as MATLAB, Octave, R-Lab, etc.
- It has many user-friendly, efficient, and easy-to-use functions that help to solve problems like numerical integration, interpolation, optimization, linear algebra, and statistics.
- The benefit of using the SciPy library in Python while making ML models is that it makes a strong programming language available for developing fewer complex programs and applications.

Installation of SciPy

To install SciPy in your system, you can use Python package manager pip. Before proceeding, make sure that you have Python already installed in your system. Here's the step to install Python in your system.

Step 1: Firstly, Open terminal and Command Prompt in your system.

Step 2: Run the installation Command to install SciPy in your system.

```
pip install scipy
```

Step 3: Pip will be download and install SciPy along with dependencies. This process will may take some time depends on internet connection.

Step 4: To verify installation, you need to import SciPy in a Python script or interactive shell.

```
import scipy print(scipy.__version__)
```

Now the installation of SciPy is successfully completed.

5) Matplotlib:

- Matplotlib is a powerful and versatile open-source plotting library for Python, designed to help users visualize data in a variety of formats.
- Developed by John D. Hunter in 2003, it enables users to graphically represent data, facilitating easier analysis and understanding.

- Matplotlib is a widely used and comprehensive plotting library in Python, primarily utilized for creating static, animated, and interactive visualizations.
- It provides a robust set of tools for generating high-quality figures and a wide variety of plot types, making it a cornerstone for data analysis, scientific research, and visual communication in Python.
- If you want to convert your boring data into interactive plots and graphs, Matplotlib is the tool for you.

Matplotlib Pyplot

Pyplot is a module within Matplotlib that provides a MATLAB-like interface for making plots. It simplifies the process of adding plot elements such as lines, images, and text to the axes of the current figure. **Steps to Use Pyplot:**

- **Import Matplotlib:** Start by importing matplotlib.pyplot as plt.
- **Create Data:** Prepare your data in the form of lists or arrays.
- **Plot Data:** Use plt.plot() to create the plot.
- **Customize Plot:** Add titles, labels, and other elements using methods like plt.title(), plt.xlabel(), and plt.ylabel().
- **Display Plot:** Use plt.show() to display the plot.

Programs:

1) Statistics:

```
import statistics as stats
data = [2, 4, 6, 8, 10, 12, 14]
# Mean
mean = stats.mean(data)
print(f"Mean: {mean}")
# Median
median = stats.median(data)
print(f"Median: {median}")
# Mode
data_with_mode = [2, 2, 3, 3, 3, 4, 5]
mode = stats.mode(data_with_mode)
print(f"Mode: {mode}")
# Standard Deviation
std_dev = stats.stdev(data)
print(f"Standard Deviation: {std_dev}")
# Variance
variance = stats.variance(data)
print(f"Variance: {variance}")
```

Output:

```
Mean: 8
Median: 8
Mode: 3
Standard Deviation: 4.320493798938574
Variance: 18.666666666666668
```

2) Maths

```
import math
# Basic mathematical functions
x = 16
sqrt = math.sqrt(x)
print(f"Square Root of {x}: {sqrt}")
# Trigonometric functions
angle = math.radians(45)
sin_angle = math.sin(angle)
print(f"Sine of 45 degrees: {sin_angle}")
# Exponential and logarithmic functions
log_value = math.log(100, 10)
print(f"Logarithm base 10 of 100: {log_value}")
exp_value = math.exp(2)
print(f"e^2: {exp_value}")
```

Output:

```
Square Root of 16: 4.0
Sine of 45 degrees: 0.7071067811865475
Logarithm base 10 of 100: 2.0
e^2: 7.38905609893065
```

3) Numpy:

```
import numpy as np

x = np.array([1, 2, 3])
y = np.array([4, 5, 6])

# Addition
add = x + y
print("Addition:", add)

# Subtraction
subtract = x - y
print("subtraction:", subtract)

# Multiplication
multiply = x * y
print("multiplication:", multiply)

# Division
divide = x / y
print("division:", divide)
```

Output:

Addition: [5 7 9]
subtraction: [-3 -3 -3]
multiplication: [4 10 18]
division: [0.25 0.4 0.5]

4) Scipy:

```
from scipy import stats as scipy_stats
import numpy as np
# Generate a random sample
data = np.random.normal(loc=0, scale=1, size=1000)
# Descriptive statistics
mean = np.mean(data)
std_dev = np.std(data)
print(f"Mean: {mean}, Standard Deviation: {std_dev}")
# T-test
t_stat, p_value = scipy_stats.ttest_1samp(data, 0)
print(f"T-statistic: {t_stat}, P-value: {p_value}")
# Pearson correlation coefficient
array1 = np.array([1, 2, 3, 4, 5])
array2 = np.array([6, 7, 8, 9, 10])
correlation, _ = scipy_stats.pearsonr(array1, array2)
print(f"Pearson Correlation Coefficient: {correlation}")
```

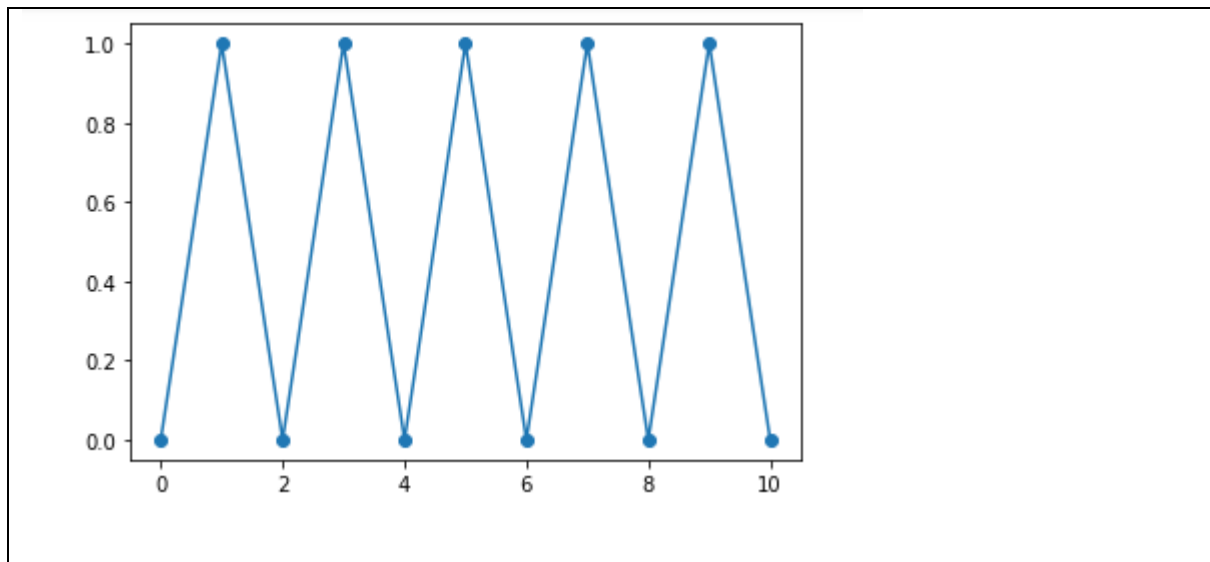
Output:
Mean: 0.046713519493229314,
Standard Deviation: 0.9970317141865672
T-statistic: 1.4808680404694945,
P-value: 0.1389570508379236
Pearson Correlation Coefficient: 1.0

5) matplotlib:

```
import matplotlib.pyplot as plt

x = [0,1,2,3,4,5,6,7,8,9,10]
y = [0,1,0,1,0,1,0,1,0,1,0]
plt.plot(x, y, marker="o")
plt.show()
```

Output:



Conclusion: This experiment, I gained a foundational understanding of essential Python libraries used in scientific and analytical computing.